Problem Summary

Merge two binary trees into a new tree.

Rules:

- If both nodes exist → sum values
- If one node exist → use its values
- Build a completely new tree

Key Idea

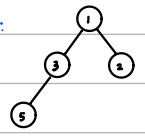
Instead of modifying root1, we create a new node at each step.

Value = (root1 value if exists) + (root2 value if exists)

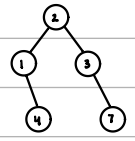
Then recursively build left and right

Diagram Examples

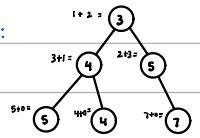
Tree 1 :



Tree 2:



Merged Tree:



0 → when root is None

Approach

1) If both nodes are None → return None
2) Compute value : Val = (root1.val if exists) + (root2.val if exists)
3) Create new node with val
4) Recursively build left and right child
5) Return new node

Time Complexity : $O(n_1 + n_2)$ → Each node is visted once in both trees

Space Complexity : $O(\max(h_1, h_2))$ → Recursion stack, where $\max(h_1, h_2)$ represents the height of the tree larger in terms of height